

The quantum number that describes the orientation of an orbital in space is:

- A. Principle quantum number
- B. Magnetic quantum number
- C. Secondary(angular momentum) quantum number
- D. Spin quantum number
- E. None of the above



Select one:

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E

B

The molar mass of compound $S_2O_x = 112 \text{ g/mol}$

Atomic weights: S = 32, O = 16 g/mol

What is the value x?

A. 3

B. 5

C. 7

D. 4

E. 6

A

Select one:

Which element has the greatest electron affinity (EA) for ^{14}Si , ^{11}Na , ^{18}Ar

- A. S
- B. Na
- C. Ar

Select one:

- ☐ A
- ☐ B
- ☐ C

Si

GENERAL CHEMISTRY 1-1

GENERAL CHEMISTRY 1-1 ► General ► MIDTERM EXAM

The element X (atomic number = 30) is a _____ element with _____ unpaired electrons

- A. paramagnetic , 3
- B. paramagnetic , 2
- C. paramagnetic, 6
- D. diamagnetic , 2
- E. diamagnetic , 0

E

Select one:

☐ A

Find the empirical formula for a compound that contains 82.8% of "C" and 17.2% of "H"

(Atomic weights: C = 12 g/mol, H = 1 g/mol)

- A. C_2H_5
- B. C_3H_8
- C. C_2H_3
- D. C_4H_{11}
- E. C_3H_5

A

Select one:

☐ A

6

answered
out of 1.50
question

How many moles is 40g of Ca?
(Atomic weight of Ca = 40 g/mol)

- A. 1.0 mole
- B. 0.5 mole
- C. 2.5 mole
- D. 2.0 mole
- E. 1.5 mole

A

Select one:

- ☒ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E

n7
answered
out of 1.50
question

Calculate the mass percentage of oxygen (O) in
of $\text{Al}(\text{NO}_3)_3$
(Atomic weights: Al =27 , N=14, O =16 g/mol)

- A. 17.1%
- B. 19.7%
- C. 58.5%
- D. 67.6%
- E. 12.7 %

D

Select one:

- ☐ A
- ☐ B
- ☐ C

For the equation: $2\text{C}_3\text{H}_7\text{OH} + 9\text{O}_2 \rightarrow 6\text{CO}_2 + 8\text{H}_2\text{O}$

If 12 g $\text{C}_3\text{H}_7\text{OH}$ (60 g/mol) are allowed to react with 20 g O_2 (32g/mol)

The mass of CO_2 (44g/mol) produced in grams is:

- A. 13.75
- B. 18.33
- C. 22.91
- D. 26.4
- E. 17.6

B

Select one:

- ☐ A
- ☐ B
- ☐ C
- ☐ D

n 9

answered

out of 1.50

question

Which electron in the nitrogen ground state atom has the quantum number set (n, ℓ, m_ℓ, m_s) equal to $(2, 0, 0, \frac{1}{2})$

(Atomic no. of N = 7)

- A. first electron
- B. fourth electron
- C. fifth electron
- D. sixth electron
- E. seventh electron

D

Select one:

- ☐ A
- ☐ B

General Chemistry II - Official Midterm Exam

The principal quantum number (n) of the first s-orbital is ____.

- A. 1
- B. 2
- C. 3
- D. 4
- E. 0

A

Select one:

- ☐ a. D
- ☐ b. A
- ☐ c. B
- ☐ d. C

How many electrons are there in the valence shell of the O^- ion?

(atomic number =8)

A. 2

B. 8

C. 7

D. 5

E. 4

Select one:

☐ A

☐ B

☐ C

☐ D

1s² 2s² 2p⁶ 3s² 3p⁵ is a ground state configuration for a

- A. Transition element
- B. Alkali metal element
- C. Representative element
- D. Inner transition element
- E. Noble element

Select one:

- ☐ A
- ☐ B
- ☒ C
- ☐ D
- ☐ E

C

Question 14

Not yet answered

Marked out of 1.50

Flag question

Which of the elements below, is a metal

- A) Na (atomic no. = 11)
- B) Si (atomic no. = 14)
- C) B (atomic no. = 5)
- D) Cl (atomic no. = 17)
- E) Ar (atomic no. = 18)

Select one:

- ☐ A
- ☐ B
- ☐ C
- ☐ D

A

How many atoms (total number of atoms) are found
in 7.5 mole of $K_2Cr_2O_7$

Avogadro's number = 6.022×10^{23}

- A. 2.3×10^{25}
- B. 7×10^{25}
- C. 5×10^{25}
- D. 1.4×10^{26}
- E. 3.6×10^{24}

C

Select one:

Question 16

Answered

Score: 1.50

Question

Which element has the greatest IE in period 2?

- A. Ne
- B. Be
- C. Li
- D. N

A

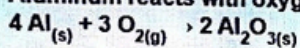
Select one:

- ☐ A
- ☐ B
- ☐ C
- ☐ D

GENERAL CHEMISTRY 1-1

GENERAL CHEMISTRY 1-1 ► General ► MIDTERM EXAM

Aluminum reacts with oxygen to produce aluminum oxide.



If 6.0 moles of Al react with excess O_2 , how many moles of Al_2O_3 can be formed?

- A. 1.5 mol
- B. 2.0 mol
- C. 2.5 mol
- D. 5.0 mol
- E. 3.0 mol

E

Select one:

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E

01-GENERAL CHEMISTRY 1-1

2021/2020-3-30206101-GENERAL CHEMISTRY 1-1 ▶ General ▶ MIDTERM EXAM

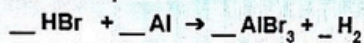
Question 18

Not yet answered

Marked out of 1.50

Flag question

For the equation:



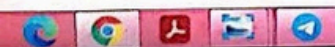
The coefficients in the balanced equation (respectively) are:

- A. 2,6,2,3
- B. 6,2,3,2
- C. 2,6,3,2
- D. 6,2,2,3
- E. 3,6,2,3

D

Select one:

- ☐ A
- ☐ B



For the equation: $2 \text{ZnS} + 3 \text{O}_2 \rightarrow 2 \text{ZnO} + 2 \text{SO}_2$

If 3.6 mol ZnS are allowed to react with excess amount of O_2
Calculate percentage yield of the reaction given that 100 g of SO_2 (64g/mol) is actually produced :

- A. 86.8 %
- B. 60 %
- C. 43.4%
- D. 26 %
- E. 30 %

C

Select one:

- ☐ A
- ☐ B
- ☐ C

Which of the following, is the atomic number for an element that belongs to period 5, group 5B (VB) in the periodic table?

A. 73

B. 42

C. 41

D. 44

E. 43

C

4

Select one:

☐ A

☐ B

☐ C

☐ D

☐ E